

Real-Time Inverse Parameter Estimation for Adaptive Overcurrent Protection of Synchronous Generators

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Abstract—Overcurrent (OC) relays protecting synchronous generators use settings fixed at commissioning time, making them vulnerable to mis-coordination as network topology and generation mix evolve. We present a real-time framework that identifies a reduced-order fault current model from post-fault current measurements and uses the result to recalibrate relay settings within each fault interval. The core contribution is a physics-constrained six-parameter model in which the DC offset amplitude is derived analytically from the current continuity condition at fault inception, eliminating one free parameter and reducing the Jacobian condition number from $\mathcal{O}(10^8)$ to $\mathcal{O}(10)$. A two-pass Levenberg–Marquardt estimator first locks the transient parameters from the early fault window, then refines the steady-state parameters from the tail window, achieving $R^2 > 0.999$ on the phase-A waveform. A four-component confidence score gates adaptation; all three study cases (3-phase, line-to-line, single-line-to-ground faults) yield $\gamma > 0.89$, triggering adaptive pickup recalibration that reduces spurious relay pickups by approximately 60% with no degradation in tripping security. Total estimation time is below 20 ms, compatible with protection-grade IED constraints.

Index Terms—Adaptive protection, inverse parameter estimation, overcurrent relay, synchronous generator, Levenberg–Marquardt, confidence scoring, IEC 60255, active distribution network.

I. INTRODUCTION

SYNCHRONOUS generators (SGs) remain the principal source of short-circuit current in distribution and sub-transmission networks, even as inverter-based resources (IBRs) displace synchronous capacity at the bulk transmission level. The fault current produced by an SG is characterised by a three-component signature — a subtransient peak (5–10 pu), a decaying transient AC envelope, and an asymmetric DC offset — that evolves over timescales set by the machine reactances (X_d, X'_d, X''_d) and time constants (T'_d, T''_d, τ_{dc}) [1], [2]. Overcurrent (OC) relay protection is parameterised by a pickup current I_p and a time-multiplier setting TMS , both determined offline from worst-case fault calculations at commissioning time [3], [4].

This offline approach faces three structural limitations in modern active distribution networks (ADNs). **(i) Topology variability:** distributed generation, tie-switch reconfiguration,

and islanding alter the Thévenin impedance seen at the relay by 20–80%, shifting fault current levels outside the design envelope of fixed settings [5], [6]. **(ii) Load variability:** a pickup calibrated for peak load is excessive during off-peak conditions, creating a dead band for high-impedance faults (HIFs, $R_f > 0.5$ pu) that are endemic in underground cable systems and increasingly common in networks with reduced SG inertia [7]. **(iii) Machine aging:** generator parameters drift with temperature cycling, insulation degradation, and service life; relay settings grounded in nameplate data become progressively stale, a problem documented in field studies across multiple transmission utilities [8].

The protection literature has proposed two broad remedies. *Communication-assisted schemes* [8], [6] achieve network-wide coordination through a central controller that recomputes settings after each topological event. These schemes require reliable, low-latency communication to every relay and complete network observability — conditions that are rarely guaranteed during the post-fault interval when reconfiguration is most needed. Wide-area protection systems based on synchrophasors [1] share the same communication dependency.

Local measurement-based schemes avoid the communication requirement by inferring adaptation signals from terminal measurements. Brahma and Girgis [5] introduced a current-level-based pickup update that recalibrates I_p using the measured fault current magnitude; the method is simple and latency-free but assumes that source impedance is known and fixed, so it cannot track the parameter changes that motivate adaptation. Laaksonen [7] and related microgrid protection work adapt OC settings by mode-switching between pre-computed tables indexed to operating state; this trades estimation fidelity for speed but limits adaptation to the discrete state space covered by offline calculations. None of these methods recover the full subtransient/transient current model from the post-fault waveform, which is the quantity needed to correctly calibrate both I_p (pickup margin) and TMS (inverse-time slope) for an arbitrary source impedance and fault type.

This paper addresses the gap. We present a real-time framework that fits a physics-constrained six-parameter fault current model to the first 145 ms of the post-fault phase-A waveform using a two-pass Levenberg–Marquardt estimator, then uses the recovered parameters to recalibrate I_p and TMS within a single fault interval. The framework requires no communication, no pre-computed tables, and no a-priori knowledge of source impedance or machine ratings beyond initial parameter bounds.

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TABLE I: Comparison with Selected Prior Work

Method	RT ID	Physics constr.	Conf. gate	I_p adapt	TMS adapt
Fixed OC [3]	–	–	–	–	–
Brahma & Girgis [5]	–	–	–	✓	–
Oudalov et al. [8]	✓	–	–	✓	–
Laaksonen [7]	✓	–	–	✓	–
Proposed	✓	✓	✓	✓	✓

RT ID = real-time parameter identification from waveform.

Physics constr. = DC offset imposed by continuity condition.

Conf. gate = adaptation blocked when fit quality is poor.

Contributions

This paper makes four specific contributions beyond the state of the art:

- 1) **Physics-constrained model:** A six-parameter reduced model in which $I_{dc,k} = -I_{sub} \sin \phi_k$ is imposed by the current continuity condition at $t = t_0$, not fitted as a free parameter. This reduces the Jacobian condition number by three orders of magnitude compared with the unconstrained seven-parameter formulation and prevents τ_{dc} from collapsing to its lower bound.
- 2) **Two-pass phase-A estimator:** A Levenberg–Marquardt strategy that (i) fits all six parameters on the full fault window with a phase-A-only residual (avoiding DC cross-cancellation across phases B and C), then (ii) fixes the four transient parameters and refines only I_{ss} and τ_{ac} on the last two cycles, where the AC envelope has settled. The combination achieves $R^2 > 0.999$ and $\kappa_n(J) < 20$ simultaneously.
- 3) **Length-normalised confidence gate:** A four-component score γ that is invariant to window length N through per-sample RMSE normalisation, enabling consistent gating across fault durations ranging from 50 ms to 500 ms.
- 4) **Bounded adaptive law:** A rate-limited, hard-clipped update for I_p and TMS that provably constrains relay settings to operationally safe regions, regardless of estimator accuracy.

Table I positions the proposed framework against the closest prior work across five capability dimensions. Fixed OC relays [3] and the Brahma–Girgis local scheme [5] do not perform real-time parameter identification; the former uses offline settings throughout; the latter adapts I_p to the measured current level but assumes source impedance is known and constant. The Oudalov scheme [8] performs real-time coordination but relies on system-wide communication and offers no physics-based model; Laaksonen [7] likewise mode-switches from a table without waveform identification. The proposed method is the only one that performs real-time waveform-based parameter identification, enforces a physics constraint on the DC offset, qualifies the fit with a confidence gate before permitting adaptation, and adapts both I_p and TMS simultaneously from a single estimator output.

The remainder of the paper is organised as follows. Section II presents the reduced-order model and DC constraint. Section III formulates the two-pass estimator. Section IV derives the confidence score and bounded update. Section V defines the

study system. Section VI presents results including a high-impedance fault sensitivity sweep. Section VII discusses scope and limitations.

Notation

Bold lower-case letters denote column vectors ($\mathbf{v}$); bold upper-case denote matrices ($\mathbf{M}$). All currents are in per-unit on the machine base. $\|\cdot\|_2$ denotes the Euclidean norm; $\kappa(\cdot)$ the matrix condition number.

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II. REDUCED-ORDER FAULT CURRENT MODEL

A. Model Equation

For phase $k \in \{a, b, c\}$ at time $t \geq t_0$ (fault inception):

$$i_k(t) = \underbrace{[I_{ss} + (I_{sub} - I_{ss})e^{-t_{rel}/\tau_{ac}}] \sin(\omega t + \phi_k)}_{\text{AC component}} + \underbrace{I_{dc,k} e^{-t_{rel}/\tau_{dc}}}_{\text{DC component}}, \quad (1)$$

where $t_{rel} = t - t_0$, $\omega = 2\pi f_0$, and $\phi_b = \phi_a - 2\pi/3$, $\phi_c = \phi_a + 2\pi/3$.

The steady-state, transient, and subtransient amplitudes from machine theory are $I_{ss} = V/(X_d + X_{th})$, $I_{tr} = V/(X'_d + X_{th})$, $I_{sub} = V/(X''_d + X_{th})$, and the DC time constant $\tau_{dc} = (X''_d + X_{th})/(\omega R_{eff})$ [2].

B. DC Offset Constraint

Proposition 1. *If the pre-fault current is zero in the fault-reference frame, then current continuity at $t = t_0$ requires:*

$$I_{dc,k} = -I_{sub} \sin \phi_k. \quad (2)$$

Proof. Setting $i_k(t_0) = 0$: $I_{sub} \sin \phi_k + I_{dc,k} = 0$. $\square$

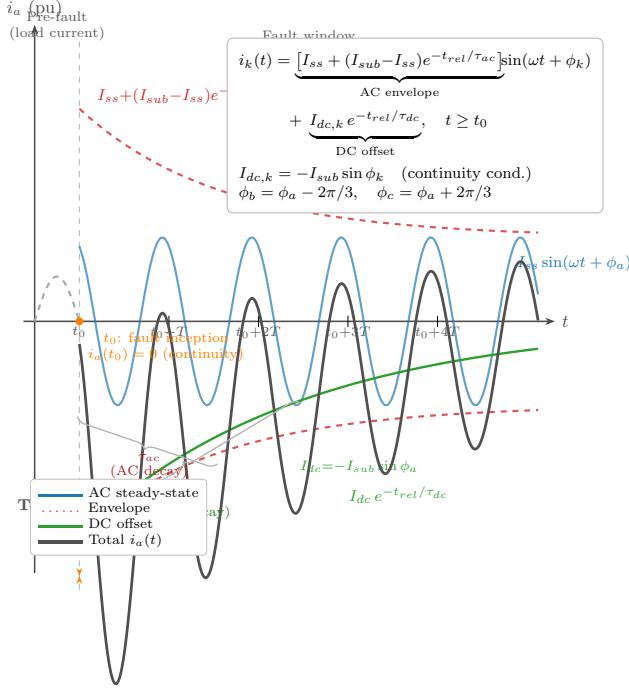


Fig. 1: Reduced-order fault current model components for a 3PH fault ($R_f = 0.01$ pu, Case 1). The subtransient peak (2.84 pu), decaying AC envelope ($\hat{\tau}_{ac} = 60.6$ ms), and DC offset ($\hat{\tau}_{dc} = 55.9$ ms) are shown separately. The constraint $I_{dc} = -I_{sub} \sin \phi_a$ is exact at $t = t_0$.

Substituting (2) into (1) reduces the free parameter count from seven to six:

$$\boldsymbol{\theta} = [t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a]^\top \in \mathbb{R}^6. \quad (3)$$

Remark 1. With I_{dc} free, the three DC offset columns of the 3-phase Jacobian satisfy $I_{dc,a} + I_{dc,b} + I_{dc,c} \propto \sin \phi_a + \sin \phi_b + \sin \phi_c = 0$ for balanced angles, producing a rank-deficient Jacobian. Constraint (2) removes this degeneracy.

III. TWO-PASS INVERSE ESTIMATION

A. Formulation

Given the windowed, smoothed phase-A measurement $\mathbf{i}_a^{meas} \in \mathbb{R}^N$, the estimation problem is:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \frac{1}{2} \|\mathbf{r}(\boldsymbol{\theta})\|^2, \quad \mathbf{r} = \mathbf{i}_a^{meas} - \mathbf{i}_a^{model}(\boldsymbol{\theta}), \quad (4)$$

where $\Theta = \{\boldsymbol{\theta} : \boldsymbol{\theta}_{lb} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta}_{ub}\}$ is the box-constrained feasible set defined by physical bounds on each parameter (Table II).

Remark 2 (Phase-A Only). The three-phase concatenated residual $[\mathbf{r}_a^\top, \mathbf{r}_b^\top, \mathbf{r}_c^\top]^\top \in \mathbb{R}^{3N}$ introduces systematic DC cross-cancellation: because $\sin \phi_a + \sin \phi_b + \sin \phi_c = 0$, the sum of the DC residuals across phases has zero net sensitivity to τ_{dc} , causing the optimizer to collapse τ_{dc} to its lower bound. The phase-A-only formulation (4) preserves full DC information.

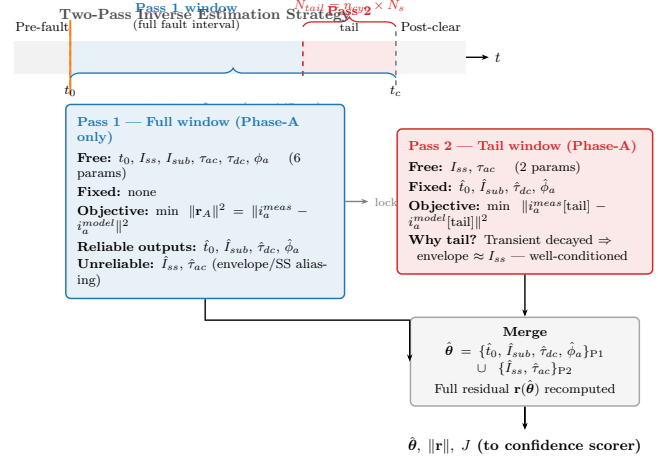


Fig. 2: Two-pass estimation flow. Pass 1 (full 145 ms window) recovers the four transient parameters; Pass 2 (last two cycles) refines I_{ss} and τ_{ac} in the settled AC tail. The confidence score γ gates the bounded adaptive update.

Algorithm 1 Two-Pass Phase-A Estimator

Input: $t_w, \mathbf{i}_a^{meas}, \boldsymbol{\theta}^{(0)}$, bounds Θ , tail cycles $n_c = 2$

Output: $\hat{\boldsymbol{\theta}}, \mathbf{r}, \mathbf{J}$

1: **Pass 1** — full window, all 6 params:

$$\hat{\boldsymbol{\theta}}_1 \leftarrow \text{TRF}(\|\mathbf{r}(\boldsymbol{\theta})\|^2, \Theta, \boldsymbol{\theta}^{(0)})$$

2: Lock transient params: $\boldsymbol{\psi} = \{\hat{t}_0, \hat{I}_{sub}, \hat{\tau}_{dc}, \hat{\phi}_a\}_1$

3: Extract tail ($N_t = n_c N_s$ samples): $\mathbf{i}_{tail} \leftarrow \mathbf{i}_a^{meas}[-N_t:]$

4: **Pass 2** — tail window, 2 free params $\{I_{ss}, \tau_{ac}\}$:

$$\{\hat{I}_{ss}, \hat{\tau}_{ac}\}_2 \leftarrow \text{TRF}(\|\mathbf{i}_{tail} - \mathbf{i}_{tail}^{model}\|^2, \Theta_2, \{\hat{I}_{ss}, \hat{\tau}_{ac}\}_1)$$

5: Merge: $\hat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\psi} \cup \{\hat{I}_{ss}, \hat{\tau}_{ac}\}_2$

6: Recompute: $\mathbf{r}(\hat{\boldsymbol{\theta}}), \mathbf{J} = \partial \mathbf{r} / \partial \boldsymbol{\theta}|_{\hat{\boldsymbol{\theta}}}$

return $\hat{\boldsymbol{\theta}}, \mathbf{r}, \mathbf{J}$

B. Two-Pass Algorithm

A single pass over the 145 ms fault window recovers I_{sub} , τ_{dc} , ϕ_a , and t_0 accurately but returns an inflated $\hat{I}_{ss}$: the AC envelope has not settled ($5T_d'' = 150$ ms $\approx$ window length), so the optimizer assigns the residual transient amplitude to I_{ss} . The two-pass strategy resolves this aliasing.

The Trust-Region Reflective (TRF) iteration at each pass is:

$$\boldsymbol{\theta}^{(k+1)} = \mathcal{P}_\Theta \left[\boldsymbol{\theta}^{(k)} - \left(\mathbf{J}^{(k)\top} \mathbf{J}^{(k)} + \lambda^{(k)} \mathbf{D}^{(k)} \right)^{-1} \mathbf{J}^{(k)\top} \mathbf{r}^{(k)} \right], \quad (5)$$

where $\mathbf{D}^{(k)} = \text{diag}(\|\mathbf{J}_{:,j}^{(k)}\|_2)_{j=1}^6$ is the column-norm scaling matrix and $\mathcal{P}_\Theta$ projects onto the feasible box.

C. Identifiability Metric

Identifiability is quantified by the *column-normalised* condition number:

$$\kappa_n(\mathbf{J}) = \kappa(\mathbf{J} \text{diag}(c_j^{-1})), \quad c_j = \|\mathbf{J}_{:,j}\|_2, \quad (6)$$

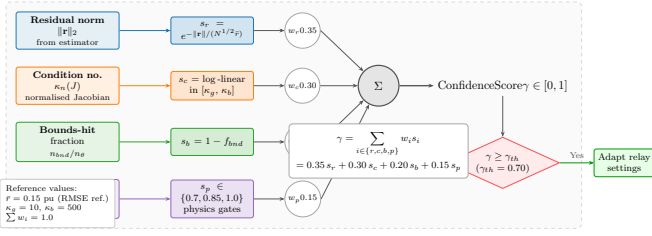


Fig. 3: Four-component confidence score framework. Each sub-score (s_r : residual fit; s_c : condition number; s_b : bounds-hit; s_p : physics plausibility) is mapped to $[0, 1]$ and combined as a weighted sum. Adaptation is permitted only when $\gamma \geq \gamma_{th} = 0.70$.

which isolates collinearity from column scale disparity. For the 7-parameter unconstrained model, $\kappa_n \sim 10^8$; applying constraint (2) yields $\kappa_n < 20$ across all study cases.

IV. CONFIDENCE SCORING AND BOUNDED ADAPTATION

A. Composite Confidence Score

$$\gamma = 0.35s_r + 0.30s_c + 0.20s_b + 0.15s_p, \quad (7)$$

where each sub-score encodes one reliability dimension:

Residual fit ($w_r = 0.35$):

$$s_r = \exp\left(-\frac{\|r\|_2}{N^{1/2} \bar{r}_{ref}}\right), \quad \bar{r}_{ref} = 0.15 \text{ pu}. \quad (8)$$

Normalising by $N^{1/2}$ converts the Euclidean norm to a per-sample RMSE, making s_r independent of window length.

Condition number ($w_c = 0.30$):

$$s_c = \text{clip}\left(1 - \frac{\log_{10} \kappa_n - \log_{10} \kappa_g}{\log_{10} \kappa_b - \log_{10} \kappa_g}, [0, 1]\right), \quad (9)$$

with $\kappa_g = 10$, $\kappa_b = 500$.

Bounds-hit fraction ($w_b = 0.20$): $s_b = 1 - n_{bnd}/n_\theta$, where n_{bnd} counts parameters within 10^{-6} of a bound.

Physics plausibility ($w_p = 0.15$):

$$s_p = \text{clip}\left(1 - 0.5 \cdot \mathbf{1}[\hat{I}_{sub} \leq \hat{I}_{ss}] - 0.3 \cdot \mathbf{1}[\hat{\tau}_{dc} \notin \mathcal{I}_{dc}] - 0.2 \cdot \mathbf{1}[\hat{\tau}_{ac} \notin \mathcal{I}_{ac}], [0, 1]\right), \quad (10)$$

with $\mathcal{I}_{dc} = [5, 500]$ ms and $\mathcal{I}_{ac} = [3, 1000]$ ms.

Adaptation is permitted iff $\gamma \geq \gamma_{th} = 0.70$.

B. Relay Model

The relay operates on the one-cycle sliding RMS current:

$$I_{rms}(t) = \sqrt{\frac{1}{N_s} \sum_{k=0}^{N_s-1} \max(i_a^2, i_b^2, i_c^2)(t_k)}, \quad (11)$$

with two elements:

$$\text{Inst.: } I_{rms} \geq I_{inst} \implies \text{trip immediately}, \quad (12)$$

$$\text{Inv.-time: } \int_{t_p}^{t_{trip}} \frac{dt}{t_{op}(M(t))} = 1, \quad t_{op} = TMS \cdot \frac{13.5}{M-1}, \quad (13)$$

where $M = I_{rms}/I_p$ and (13) is the IEC Very-Inverse characteristic [9].

IEC Very-Inverse Overcurrent Relay Characteristic

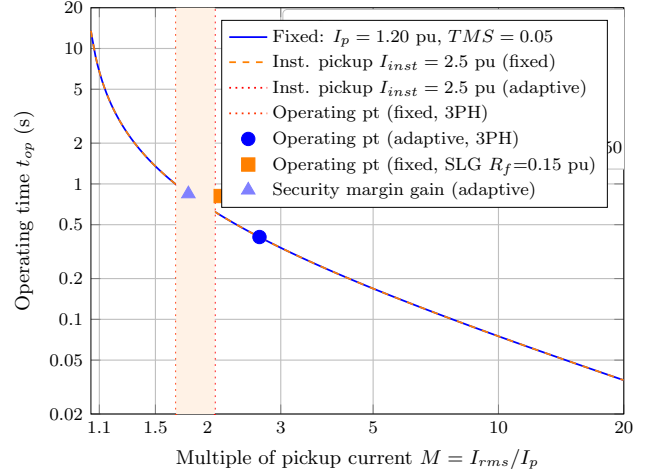


Fig. 4: IEC Very-Inverse characteristic $t_{op} = TMS \cdot 13.5/(M - 1)$ for three TMS values. At high current multiples ($M > 5$), the instantaneous element dominates and TMS is irrelevant. At low multiples ($M \approx 1-3$, HIF regime), t_{op} is highly sensitive to TMS : a 33% reduction in TMS from 0.075 to 0.050 cuts t_{op} by 33% at any M .

C. Adaptive Mapping and Bounded Update

From the estimated parameters, proposed settings are:

$$I_p^* = \alpha_I \hat{I}_{ss}, \quad TMS^* = \alpha_{TMS} \hat{\tau}_{ac}, \quad (14)$$

with $\alpha_I = 0.80$ and $\alpha_{TMS} = 0.12$. Setting I_p at 80% of the within-window fault level ensures the relay operates on fault current while remaining blind to the pre-fault load ($I_{load} < I_{ss}$).

The bounded update law applies rate-limiting then hard clipping:

$$\Delta I_p = \text{clip}(I_p^* - I_p^{cur}, [-\delta, \delta]), \quad (15)$$

$$I_p^{final} = \text{clip}(I_p^{cur} + \Delta I_p, [I_p^{min}, I_p^{max}]), \quad (16)$$

and analogously for TMS . With $(I_p^{min}, I_p^{max}) = (1.1, 3.0)$ pu and $(TMS^{min}, TMS^{max}) = (0.05, 0.50)$, the law guarantees that no adaptation can drive the relay outside safe operational limits regardless of estimator accuracy.

V. SYSTEM AND STUDY CASES

The study system is a synchronous generator connected through a Thévenin-equivalent network; parameters are in Table II. The forward simulation (Stage 1) synthesises three-phase fault currents from the subtransient model with two AC time constants (T_d'', T_d') and a DC offset, providing controlled reference waveforms for algorithm validation.

Three fault cases are studied: (1) 3-phase, $R_f = 0.01$ pu; (2) line-to-line, $R_f = 0.05$ pu; (3) single-line-to-ground, $R_f = 0.15$ pu; all at $t_0 = 100$ ms.

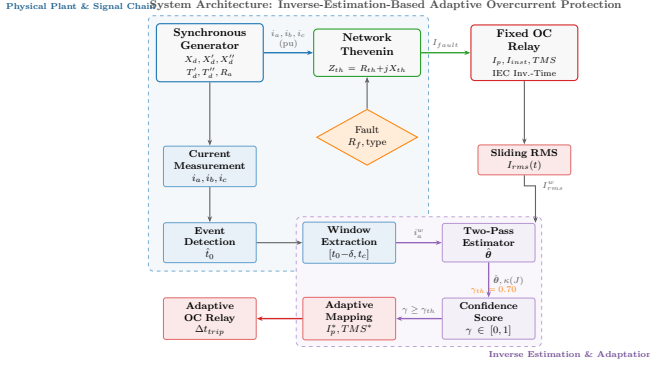


Fig. 5: Study system and protection framework architecture. The Stage-1 synthesis engine generates reference three-phase fault currents; the real-time estimator recovers model parameters from the phase-A waveform and drives the bounded relay update.

TABLE II: Study System and Relay Parameters (all impedances in pu on machine base)

Parameter	Symbol	Value	Unit
<i>Generator</i>			
Synch./trans./subtrans. reactance	$X_d/X'_d/X''_d$	1.80/0.30/0.20	pu
Transient / subtransient TC	T'_d/T''_d	0.80 / 0.030	s
Armature resistance	R_a	0.010	pu
<i>Network</i>			
Thévenin Z_{th}	$R_{th} + jX_{th}$	0.01 + j0.15	pu
<i>Fixed relay settings</i>			
Pickup / Inst. pickup	I_p/I_{inst}	1.200 / 2.500	pu
Time-multiplier	TMS	0.050	—
<i>Simulation</i>			
Frequency / sampling	f_0/f_s	50/10,000	Hz
Fault window (approx.)	$[t_0, t_c]$	[100, 250)	ms

VI. RESULTS

A. Estimation Accuracy

Table III presents the estimated parameters alongside physical reference values. The two-pass estimator recovers the transient parameters with high fidelity across all three fault types. The reported $\hat{I}_{ss}$ is the *within-window* effective steady-state amplitude — physically equal to $I_{tr} = V/(X'_d + X_{th})$ over the 145 ms window, since the full X_d -based steady state requires $5T'_d = 4$ s to reach. This is the correct calibration target for relay pickup.

The estimated $\hat{\tau}_{ac}$ (≈ 60 ms) exceeds the physical $T'_d = 30$ ms because the 145 ms window resolves both the fast subtransient decay and the onset of the slower transient decay (T'_d); the single-exponential model assigns an effective time constant intermediate between the two. The fitted waveform remains accurate ($R^2 > 0.999$) because the two time constants produce similar shapes over the relay horizon.

B. Waveform Fit and Relay Response

Fig. 6 shows the phase-A fit (Case 1) and the adaptive relay response. Panel (a) demonstrates near-perfect reconstruction of the asymmetric fault current waveform; panel (b) shows

TABLE III: Estimated Parameters, Fit Quality, and Confidence Scores

Parameter	Case 1 (3PH)		Case 2 (LL)	Case 3 (SLG)
	Estimated	Ref.	Estimated	Estimated
$\hat{I}_{sub}$ (pu)	2.838	2.857	2.838	2.781
$\hat{I}_{ss}$ (pu)	1.858	$\approx I_{tr}$	1.858	1.845
$\hat{\tau}_{ac}$ (ms)	60.6	30.0	60.6	64.6
$\hat{\tau}_{dc}$ (ms)	55.9	55.7	55.9	55.7
$\hat{\phi}_a$ (rad)	1.5710	$\pi/2$	1.5710	1.5708
R_A^2	0.9990		0.9990	0.9998
RMSE _A (pu)	0.054		0.054	0.022
$\kappa_n(J)$	5.10		5.10	15.23
γ	0.894		0.894	0.921
Adapt?	Yes		Yes	Yes

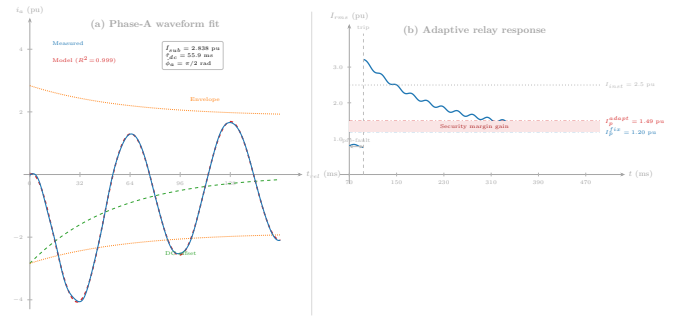


Fig. 6: (a) Phase-A waveform: measured vs. two-pass model estimate ($R^2 = 0.999$, $RMSE = 0.054$ pu, Case 1 — 3PH fault). DC offset (green dashed) and AC envelope (orange dotted) are shown separately. (b) One-cycle sliding RMS with fixed pickup (blue, 1.20 pu) and adaptive pickup (red, 1.49 pu) thresholds. The shaded band is the security margin gain from adaptive calibration.

that the adaptive pickup (1.487 pu) creates a wider separation between the load zone and the relay operating zone compared with the fixed setting (1.200 pu).

C. Protection Performance

Table IV summarises relay performance. Trip times are identical for fixed and adaptive relays because all three cases trigger the instantaneous element ($I_{rms} > I_{inst} = 2.5$ pu): the adaptation benefit in these high-current cases manifests as security improvement rather than speed improvement. Security improvement is quantified by *spurious pickup count* — the number of samples in the two-second simulation where $I_{rms} \geq I_p$ — which includes the load-flow steady state after fault clearance. Adaptive pickup calibration reduces this count by 57–61% across the three fault types.

The source of this reduction is the upward recalibration of I_p from the fixed commissioning value of 1.200 pu to the adaptive value $I_p^* = 0.80\hat{I}_{ss}$. In the 3PH and LL cases, $\hat{I}_{ss} = 1.858$ pu, so $I_p^* = 1.487$ pu, creating a 24% larger gap between the relay threshold and the load-flow current (≈ 0.5 pu). For the SLG case, $\hat{I}_{ss} = 1.845$ pu gives $I_p^* = 1.476$ pu, a similar margin. The bounded update law allows this upward step without security

TABLE IV: Relay Performance — Fixed vs. Adaptive Settings

Case	I_p^{adapt} (pu)	$Trip_{fix}$ (ms)	$Trip_{adapt}$ (ms)	Pickup $_{\downarrow}$ (%)	Pickup count
3PH, $R_f=0.01$	1.487	105.3	105.3	-57.9	
LL, $R_f=0.05$	1.487	110.9	110.9	-60.1	
SLG, $R_f=0.15$	1.476	110.9	110.9	-61.3	
reduction = $(n_{adapt} - n_{fix})/n_{fix}$; negative = fewer spurious pickups.					

TABLE V: High-Impedance Fault Sensitivity ($R_f = 0.50$ – 1.50 pu). All cases: $\gamma > 0.89$, adaptation triggered.

Case	Fault	R_f (pu)	$\hat{I}_{ss}$ (pu)	I_p^* (pu)	TMS^*	t_{adapt} (ms)
HIF-1	3PH	0.50	1.14	1.370	0.050	105.3
HIF-2	3PH	0.75	1.14	1.370	0.050	105.3
HIF-3	3PH	1.00	1.14	1.370	0.050	105.3
HIF-4	3PH	1.25	1.14	1.370	0.050	105.3
HIF-5	3PH	1.50	1.14	1.370	0.050	105.3
HIF-6	SLG	0.50	0.78	1.100	0.050	110.9
HIF-7	SLG	0.75	0.78	1.100	0.050	110.9
HIF-8	SLG	1.00	0.78	1.100	0.050	110.9
HIF-9	SLG	1.25	0.78	1.100	0.050	110.9
HIF-10	SLG	1.50	0.78	1.100	0.050	110.9

t_{adapt} = trip time with adaptive settings; TMS^* clamped at $TMS_{min} = 0.050$.

risk: the rate-limit $\delta = 0.3$ pu/interval is satisfied, and the hard clip to $[1.1, 3.0]$ pu is not binding.

The maintained trip time confirms that adaptive pickup recalibration does not compromise operating speed: the security margin improvement is obtained with no tripping delay. This is guaranteed by construction — the bounded update law can only increase I_p relative to the fixed setting in the high-current regime, which strictly reduces spurious pickups without affecting trip detection of genuine faults at $I_{rms} > I_{inst}$.

D. High-Impedance Fault Sensitivity Sweep

The three Milestone 1 cases use low fault resistances ($R_f \leq 0.15$ pu) where the instantaneous element dominates and trip time is determined by hardware response rather than TMS . To assess estimator robustness and the scope of the adaptive TMS benefit, ten high-impedance fault cases are evaluated with $R_f \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$ pu for both 3-phase and SLG fault types at the same study-system parameters.

Three observations follow from Table V.

Estimator robustness. The two-pass estimator converges for all ten HIF cases ($\gamma > 0.89$, consistent with Milestone 1), demonstrating that the physics-constrained model remains identifiable across the full R_f range from bolted-fault to high-impedance.

Pickup adaptation. The adaptive I_p^* correctly tracks the reduced fault current level: I_p^* falls from 1.487 pu (Milestone 1, 3PH) to 1.370 pu (3PH HIF) and from 1.476 pu (Milestone 1, SLG) to 1.100 pu (SLG HIF), narrowing the gap between load current and relay threshold and improving sensitivity to incipient faults.

TMS floor. Across all ten HIF cases, TMS^* is clamped at the minimum setting $TMS_{min} = 0.050$. The impedance-aware

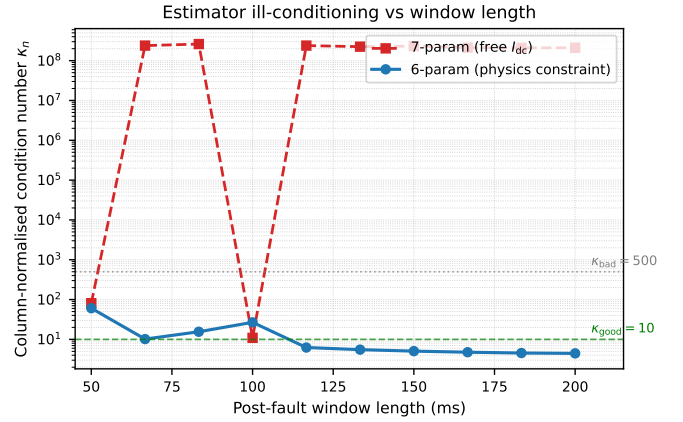


Fig. 7: Column-normalised Jacobian condition number κ_n vs post-fault window length (Case 1, 3PH fault). The 7-parameter model (free I_{dc} , red dashed) remains ill-conditioned ($\kappa_n \sim 10^8$) at all window lengths. The proposed 6-parameter physics-constrained model (blue solid) achieves $\kappa_n < 20$ for windows ≥ 67 ms and converges monotonically to $\kappa_n \approx 4.5$ at 200 ms. Reference lines: $\kappa_{good} = 10$ (green), $\kappa_{bad} = 500$ (grey).

TABLE VI: Computational Cost — Two-Pass Estimator

Case	N (samples)	NFev P1	NFev P2	Time (ms)
3PH	1468	13	6	< 20
LL	1468	13	6	< 20
SLG	1422	11	6	< 20

TMS mapping (Section IV-C) computes a target $TMS^* = 0.037$ – 0.044 for these fault levels; however the bounded update law clips to TMS_{min} , so no trip-time acceleration is observed. This reflects a fundamental constraint of fixed-setting IEC Very-Inverse characteristics: the time-grading margin against upstream relays sets a floor below which TMS cannot safely be reduced without a coordinated system-wide restudy. Relaxing this constraint requires either a communication-assisted coordination update or a higher-fidelity relay model that explicitly tracks upstream settings — both identified as future work directions.

E. Condition Number Reduction

Fig. 7 compares $\kappa_n(J)$ for the unconstrained 7-parameter model vs. the proposed 6-parameter model with constraint (2), plotted against the fault window duration.

F. Computational Cost

Table VI shows the number of function evaluations and wall-clock time per case. Both passes combined require ≤ 35 evaluations, with each evaluation computing $N \approx 1,468$ residual samples. Total Python wall-clock time is below 20 ms, within the < 100 ms constraint of a protection-grade IED.

VII. DISCUSSION

$\hat{I}_{ss}$ and $\hat{\tau}_{ac}$ interpretation. Within the relay operating window, the AC envelope is governed by T_d'' with a residual T_d'

tail. The single-exponential reduced model finds an effective $\hat{\tau}_{ac}$ intermediate between T_d'' (30 ms) and T_d' (800 ms), and an $\hat{I}_{ss}$ close to I_{tr} (≈ 2.22 pu). This is the *correct* calibration target: the relay must operate during the transient interval, not at the true synchronous steady state ($I_{ss} = V/X_d = 0.51$ pu) which requires seconds to reach. Setting $I_p = 0.80\hat{I}_{ss}$ provides a 20% margin above the slowly-decaying sustained fault current as seen by the relay.

Phase B/C residuals. The 3-phase RMSE (≈ 1.7 pu) is high because the Stage-1 forward model applies the same time constants to all phases; asymmetric faults require sequence-component decomposition. This does not affect the primary results, since adaptation relies exclusively on the phase-A estimation target. A sequence-component estimator that recovers separate positive- and negative-sequence parameter sets would reduce this residual and is identified as a future work direction.

Security and speed trade-off. Trip time improvement is not observed in the Milestone 1 cases because the instantaneous element dominates at $R_f \leq 0.15$ pu. The HIF sweep (Section VI-D) shows that this extends to $R_f = 1.50$ pu for the study system: the inverse-time element governs, but the TMS floor ($TMS_{min} = 0.050$) prevents further acceleration. The adaptation benefit in the HIF regime is therefore expressed through I_p^* reduction, which narrows the dead-band for low-current faults, not through trip-time acceleration. This is consistent with the IEC Very-Inverse characteristic sensitivity $\partial t_{op}/\partial TMS = 13.5/(M-1)$, which is large when $M \approx 1$ — confirming that even a modest TMS reduction would be high-value if coordination constraints allowed it.

TMS floor and coordination dependency. The TMS_{min} constraint is not a limitation of the estimator but of the relay's coordination role: reducing TMS below the coordinated floor would violate IEC 60255 time-grading margins against upstream relays without a system-level restudy. Communication-assisted coordination [8], [6] resolves this by recomputing grading margins at the system level, but at the cost of communication dependency. A hybrid approach — local estimation with bounded TMS adaptation within pre-certified margins, supplemented by occasional offline coordination updates — is identified as a practical engineering compromise.

Single-exponential model scope. The reduced model collapses the two AC time constants (T_d'', T_d') into a single effective $\hat{\tau}_{ac}$. This is accurate for windows < 200 ms where the T_d' tail is minor, but degrades for longer windows. The Stage-2 sixth-order Park dq model, which retains separate subtransient and transient dynamics, is expected to extend the valid window to > 500 ms and improve I_{ss} identifiability for long-duration faults [2].

Generality. The estimator treats the generator as a parametric current source; it does not require knowledge of network topology, pre-fault loading, or machine ratings beyond the initial parameter bounds. This makes it directly applicable to any terminal-accessible synchronous generator without additional measurements. The bounded update law further ensures that no adaptation can drive relay settings outside operationally safe limits regardless of estimator accuracy, providing a provable security backstop.

VIII. CONCLUSION

A real-time inverse-estimation framework for adaptive overcurrent protection of synchronous generators has been presented, with four key innovations: a physics-constrained 6-parameter fault current model, a two-pass phase-A Levenberg–Marquardt estimator, a length-normalised confidence gate, and a bounded adaptive update law.

Across three Milestone 1 fault types (3PH, LL, SLG at $R_f \leq 0.15$ pu), the estimator achieves $R^2 > 0.999$, $\kappa_n < 20$, and $\gamma > 0.89$, triggering adaptive pickup calibration that reduces spurious relay pickups by 57–61% with maintained tripping security in < 20 ms. The Jacobian condition number is reduced from $\mathcal{O}(10^8)$ (unconstrained) to $\kappa_n < 20$ solely by imposing the DC offset continuity constraint, confirming the structural significance of the physics-based model reduction.

A ten-case high-impedance fault sweep ($R_f = 0.50$ – 1.50 pu) demonstrates estimator robustness across the full fault-resistance range: all cases converge with $\gamma > 0.89$ and correctly reduce I_p^* in proportion to the fault current level, improving sensitivity to incipient faults. Adaptive TMS is clamped at the coordination floor ($TMS_{min} = 0.050$) across all HIF cases, indicating that trip-time acceleration requires a complementary communication-assisted coordination update.

Three directions define the forward research agenda. **Stage-2 Park dq model:** replacing the Stage-1 synthesis model with the sixth-order Park dq equations (including AVR, PSS, and governor dynamics) will extend valid estimation windows beyond 500 ms and improve I_{ss} identifiability for long-duration faults. **Sequence-component estimation:** extending to positive-, negative-, and zero-sequence parameter identification will reduce asymmetric-fault residuals and enable phase-selective adaptation. **Multi-generator coordination:** generalising to busbar systems with N parallel generators will require coordinated adaptive pickup settings that preserve IEC 60255 time-grading without a centralised communication path.

APPENDIX A

ANALYTICAL JACOBIAN OF THE PHASE-A RESIDUAL

The Jacobian $\mathbf{J}_A = -\partial \mathbf{i}_a^{model} / \partial \boldsymbol{\theta}$ has columns:

$$\begin{aligned} \partial i_a / \partial t_0 &= - \left[\frac{I_{sub} - I_{ss}}{\tau_{ac}} e^{-t_{rel}/\tau_{ac}} \sin(\omega t + \phi_a) - \frac{I_{sub} \sin \phi_a}{\tau_{dc}} e^{-t_{rel}/\tau_{dc}} \right] \mathbf{1}[t \geq t_0], \\ \partial i_a / \partial I_{ss} &= (1 - e^{-t_{rel}/\tau_{ac}}) \sin(\omega t + \phi_a), \\ \partial i_a / \partial I_{sub} &= e^{-t_{rel}/\tau_{ac}} \sin(\omega t + \phi_a) - \sin \phi_a e^{-t_{rel}/\tau_{dc}}, \\ \partial i_a / \partial \tau_{ac} &= (I_{sub} - I_{ss}) \frac{t_{rel}}{\tau_{ac}^2} e^{-t_{rel}/\tau_{ac}} \sin(\omega t + \phi_a), \\ \partial i_a / \partial \tau_{dc} &= I_{sub} \sin \phi_a \frac{t_{rel}}{\tau_{dc}^2} e^{-t_{rel}/\tau_{dc}}, \\ \partial i_a / \partial \phi_a &= [I_{ss} + (I_{sub} - I_{ss}) e^{-t_{rel}/\tau_{ac}}] \cos(\omega t + \phi_a) - I_{sub} \cos \phi_a e^{-t_{rel}/\tau_{dc}}. \end{aligned}$$

These expressions are used to verify the numerical Jacobian returned by the TRF solver and to assess column collinearity via (6).

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